

Paul Prae

Principal AI Engineer & Architect | Buford, Georgia, United States | hireme@paulprae.com | [LinkedIn](#) | [GitHub](#) | paulprae.com

PROFESSIONAL SUMMARY

AI engineering leader with 13+ years building production AI systems for healthcare, life sciences, and financial services. Architects enterprise solutions spanning data platforms, ML infrastructure, and cloud systems across HIPAA, FedRAMP, and GDPR-compliant environments, with deep expertise in clinical decision support, medication optimization, and neuroinformatics. Led cross-functional teams and served Fortune 500 clients at Amazon Web Services, Microsoft, Booz Allen Hamilton (\$9B+ revenue), and Slalom Consulting. Currently leading healthcare AI solution design at Autonomize AI, with hands-on experience building AI agents, data platforms, and generative AI applications on AWS and Snowflake.

PROFESSIONAL EXPERIENCE

Solutions Architect

Autonomize AI | Remote | Apr 2026 – Present

- Lead solution design and deployment for healthcare and life sciences customers adopting Autonomize AI's enterprise AI platform
- Own technical discovery, architecture design, and integration strategy across the full customer lifecycle
- Develop HIPAA-compliant solution blueprints for clinical review, risk adjustment, claims processing, and knowledge automation use cases

Staff Data Operations Engineer

Arine | Remote | Sep 2025 – Mar 2026

- Managed enterprise data platform on Snowflake and AWS, processing petabytes of healthcare data from hundreds of sources supporting Arine's medication-optimization platform (45+ health plans, >30M client members)
- Developed HIPAA-compliant AI coding assistants — a Data Engineering Agent that autonomously generated, deployed, and tested ETL pipelines, and a Dev Environment Agent that eliminated multi-day developer onboarding delays
- Built an observability pipeline monitoring Arine's autonomous pharmacist AI agent, ingesting production behavior data from DynamoDB into Snowflake and visualizing metrics in QuickSight dashboards
- Led AI enablement workshops training groups of up to 100 engineers on AI-first software development practices

Chief AI Architect, Senior Manager

Booz Allen Hamilton | Atlanta, Georgia | Jul 2024 – Mar 2025

- Led AI and analytics solution development at a Fortune 500 consulting firm (27,000+ employees, \$9B+ revenue), designing products harnessing clinical, behavioral, and genomic data for population health and predictive medicine
- Collaborated with cross-functional teams of engineers, data scientists, and domain experts to deliver AI solutions addressing algorithmic bias and data privacy risks across FedRAMP and HIPAA-compliant environments
- Contributed to AI governance policy development for FDA's Center for Devices and Radiological Health (CDRH) and supported federal healthcare AI initiatives

Chief AI Officer, Founder

Hyperbloom | United States | Jun 2021 – Aug 2025

- Grew AI consulting business to \$1.4M ARR from a \$40K investment, delivering cloud-based AI and data solutions across healthcare, life science, and financial services
- Led teams of sales coordinators, account managers, and solutions architects through the full sales cycle for enterprise AI products and services
- Architected disaster recovery solutions for a clinical-trial management platform supporting 10,000+ research sites across 45 countries

Neuroinformatics Data Platform Architect

TReNDS Center | Atlanta, Georgia | Jan 2022 – Sep 2023

- Architected COINSTAC, an open-source platform enabling decentralized neuroimaging and genomics research with differential privacy, contributing to a peer-reviewed publication in *Frontiers in Neuroinformatics*
- Built GPU-focused compute infrastructure for deep learning workloads and managed AWS cloud architecture including container orchestration and service decoupling
- Drove commercialization through SBIR grant writing, open-source contributor recruitment, and strategic industry partnerships

Enterprise AI and ML Solutions Architect

Amazon Web Services | Atlanta, Georgia | Aug 2018 – May 2021

- Managed a portfolio of 10+ enterprise accounts including Cox Enterprises, Equifax, NCR, Accenture, and Deloitte, providing AI/ML architecture guidance and driving AWS platform adoption
- Engaged nationally as a subject matter expert in AI and machine learning, authoring technical white papers, reference architectures, and delivering proof-of-concept solutions
- Partnered with C-suite executives to define technical and business strategies including go-to-market planning, cost optimization, and technical recruiting

Senior AI Engineer

NeuroLex Labs | Atlanta, Georgia | Feb 2018 – Jul 2018

- Architected automated ML pipelines for healthcare voice computing — collecting, cleaning, training, and deploying models processing text and audio data for neurodegenerative disease prediction
- Mentored fellows in the Tribe program for neurodegenerative disease research and drove test-driven development practices across the engineering team

Advanced Analytics Consultant
Slalom Consulting | Atlanta, Georgia | Jul 2015 – Jan 2018

- Delivered 7+ client engagements generating \$2M+ in consulting revenue, including a statewide behavioral health demand forecasting system across dozens of facilities in Georgia
- Designed intelligent software applications on Microsoft Azure and AWS; recognized nationally as a subject matter expert in NLP and machine learning

Software Engineer
Red Ventures | Fort Mill, South Carolina | Mar 2014 – Jul 2015

- Engineered full-stack web applications serving 16+ Fortune 500 clients including DirecTV, Verizon, AT&T, and Liberty Mutual, maintaining hundreds of high-traffic consumer-facing websites

Senior Support Engineer
Microsoft | Charlotte, North Carolina | Jul 2012 – Mar 2014

- Delivered advanced technical support to Fortune 100 clients including Chevron, Walmart, Intel, and Bank of America; nominated for Global Business Support’s “Entrepreneur of the Year” (2013)

EDUCATION

Bachelor of Science in Computer Science (AI Emphasis) & Bachelor of Arts in Cognitive Science (AI Foundation) **University of Georgia** | 2008 – 2012 Graduated Cum Laude | Dean’s List | HOPE Scholarship

TECHNICAL SKILLS

AI & Machine Learning: Generative AI, Large Language Models (LLMs), Multi-agent Systems, Natural Language Processing, Machine Learning, MLOps, Deep Learning, Prompt Engineering, Predictive Analytics

Cloud & Infrastructure: AWS (SageMaker, Bedrock, ECS/EKS, Lambda, S3, Glue, QuickSight, DynamoDB), Microsoft Azure, Snowflake, Docker, NVIDIA GPU Computing (CUDA)

Data Engineering: ETL/CDC Pipelines, Data Governance, HIPAA/GDPR/FedRAMP Compliance, PostgreSQL, Apache Kafka, Data Warehousing

Programming Languages: Python, SQL, JavaScript, C#, Bash, PowerShell

AI Tools & Platforms: Ollama, Open WebUI, Cursor AI, GitHub Copilot, Azure Machine Learning

Leadership: Engineering Team Management, Solutions Architecture, Enterprise Consulting, Executive Coaching, Agile/Scrum

CERTIFICATIONS

- **AWS Certified Machine Learning Engineer** — Amazon Web Services (Sep 2025)
- **AWS Certified AI Practitioner** — Amazon Web Services (Sep 2025)
- **AWS Certified Machine Learning - Specialty** — Amazon Web Services (Jan 2020)
- **AWS Certified Solutions Architect** — Amazon Web Services (Jul 2018)

PROJECTS

Modular Earth
Not-for-profit organization building free, open-source AI agents to help working-class entrepreneurs generate wealth. Active projects include a career assistant and a financial advisor built on local-first, privacy-preserving architectures using Ollama, open-weight LLMs, and NVIDIA GPU hardware (RTX 4090/5090). Vision: deploy AI-as-a-public-service in local libraries across the United States.

PUBLICATIONS

[Enhancing Collaborative Neuroimaging Research: Introducing COINSTAC Vaults for Federated Analysis and Reproducibility](#) Frontiers in Neuroinformatics (Jun 2023). Co-authored peer-reviewed research on federated analysis enabling multi-site neuroimaging studies without sharing raw data, using differential privacy and decentralized computation.